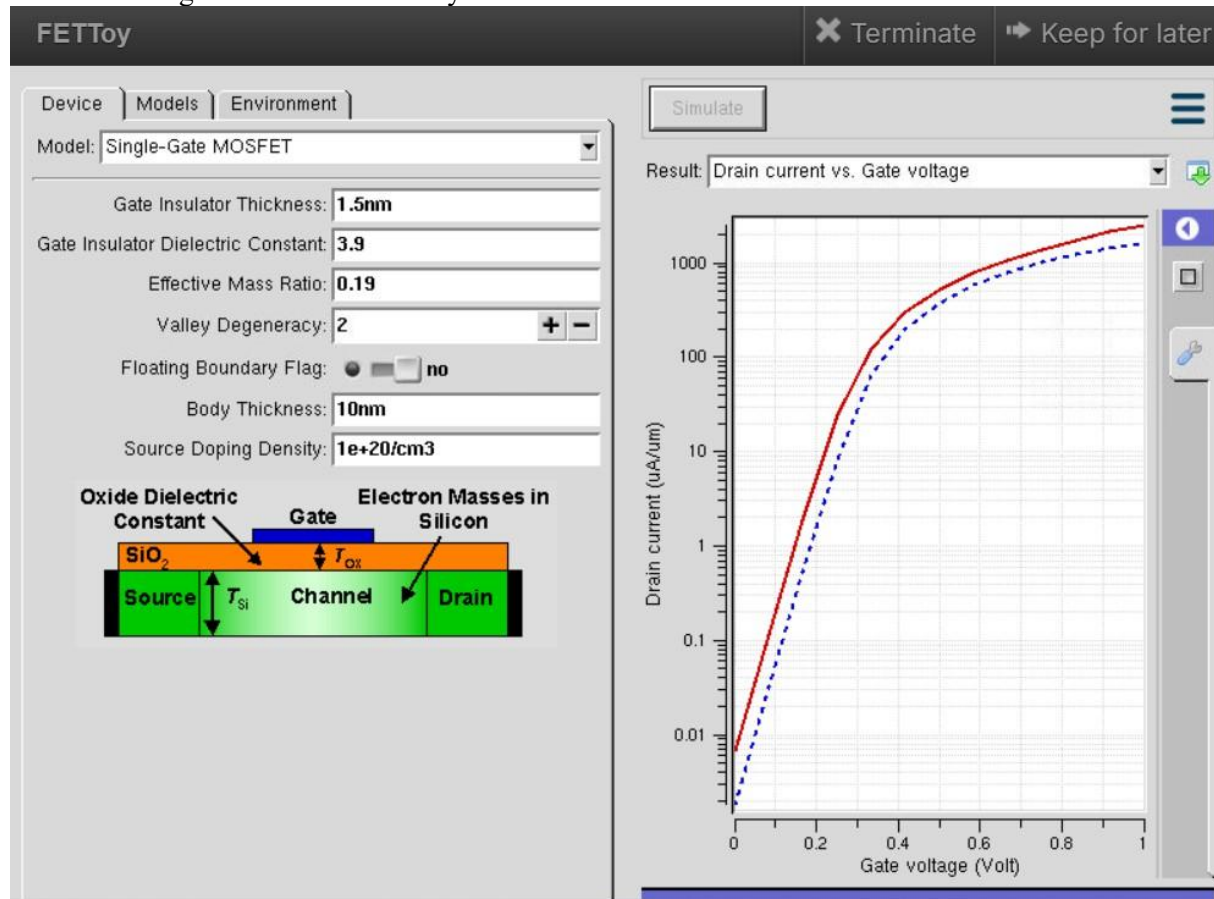


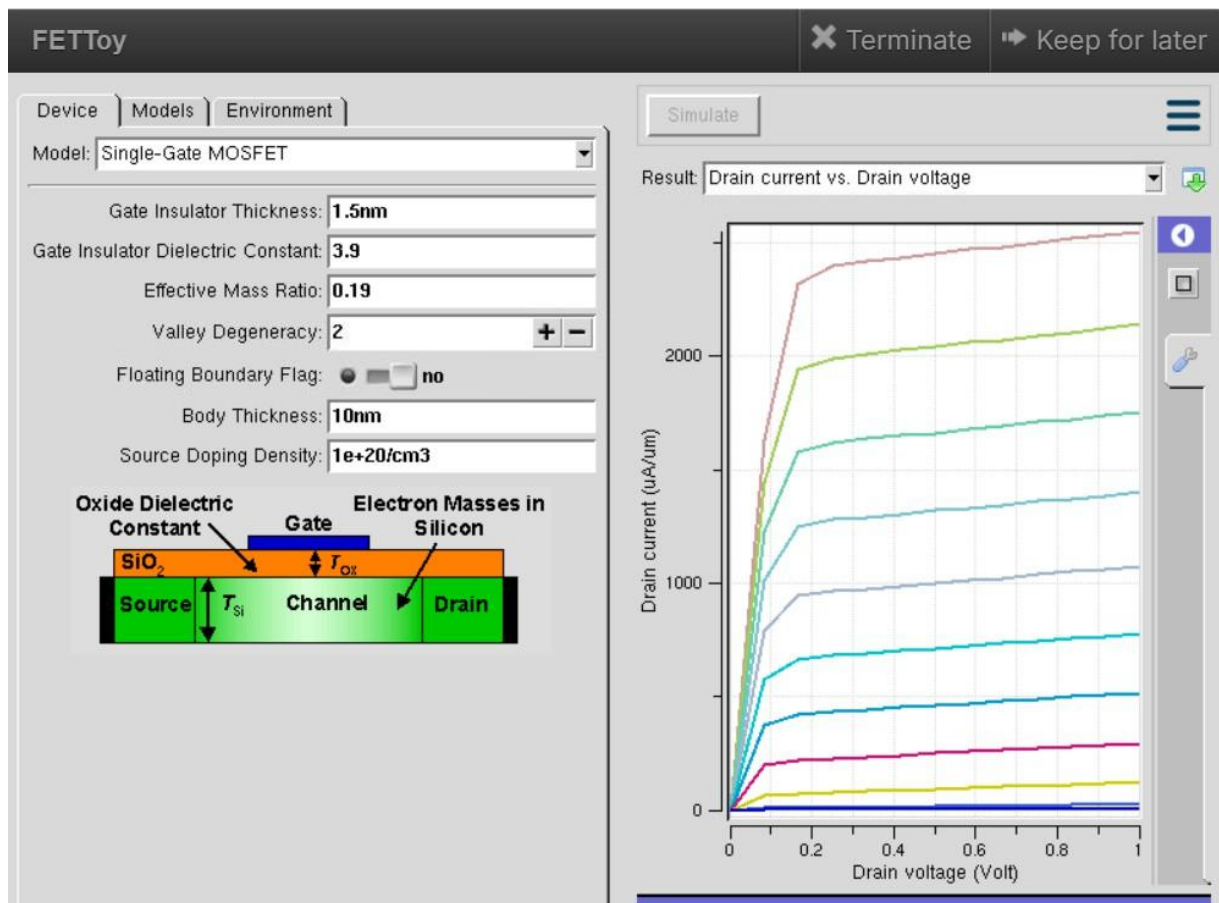
Exp. 6. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Aim:

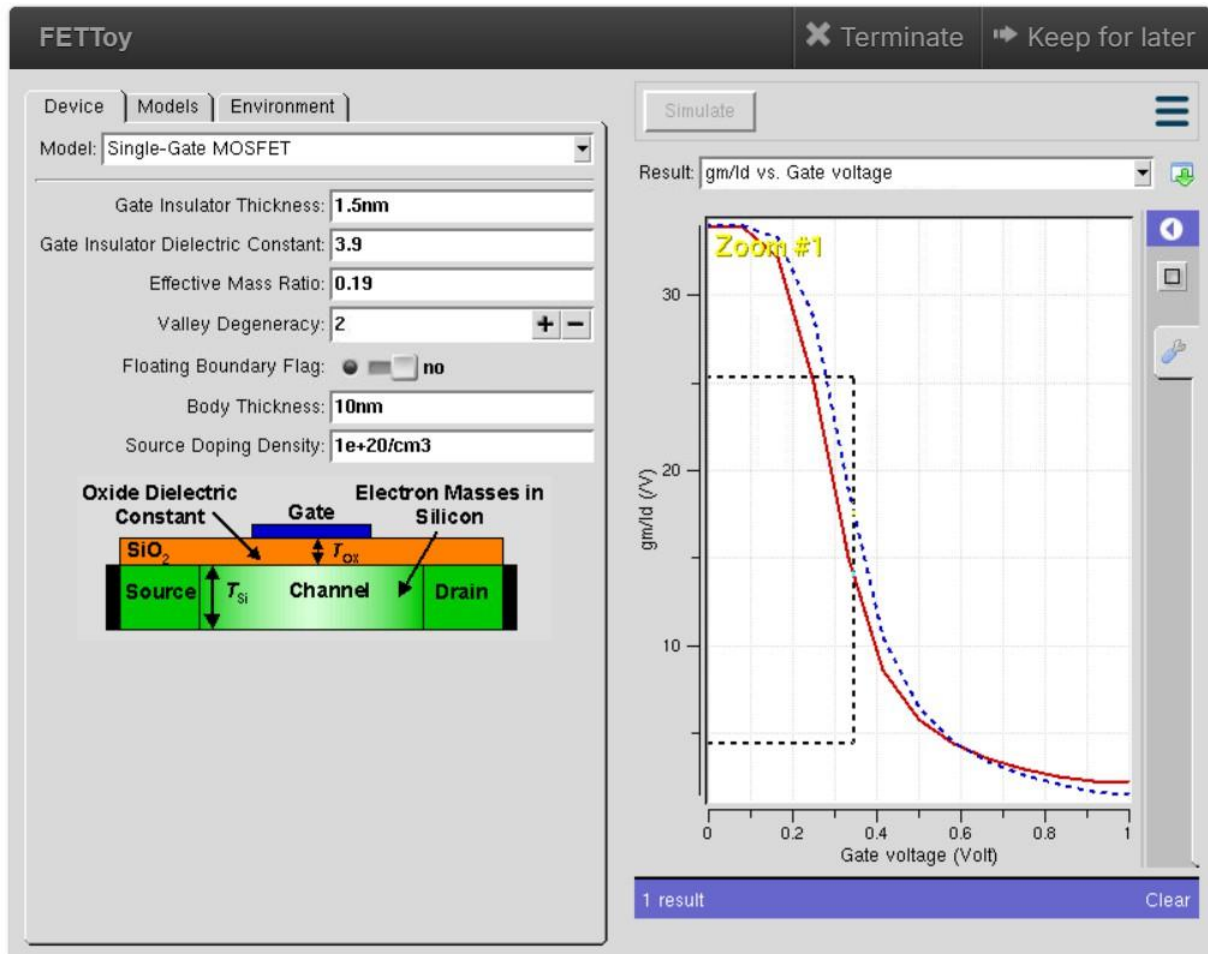
To study the current-voltage characteristics and transconductance efficiency of a single-gate MOSFET using the NanoHUB FetToy Simulator.



This image presents the FETToy simulation of a Single-Gate MOSFET, where parameters such as gate insulator thickness (1.5 nm), dielectric constant (3.9), and source doping density ($1 \times 10^{20} \text{ cm}^{-3}$) are configured. The left side illustrates the MOSFET structure consisting of the Source, Gate, Channel, Drain, and SiO₂ gate oxide, while the graph on the right shows the Drain Current (I_d) versus Gate Voltage (V_g). The graph indicates that the drain current increases significantly as the gate voltage rises, demonstrating efficient channel formation and enhanced carrier transport. This simulation is valuable for evaluating MOSFET characteristics and optimizing CMOS technology, integrated circuits, and high-performance, low-power semiconductor devices.



This image presents the FETToy simulation of a Single-Gate MOSFET, showing the relationship between Drain Current (I_d) and Drain Voltage (V_d) for different gate voltage values. The multiple output curves demonstrate that the drain current increases with drain voltage and eventually reaches a saturation region, which is a key operating characteristic of MOSFET devices. As the gate voltage increases, the channel conductivity improves, resulting in higher drain current. This simulation helps analyze MOSFET output performance and is widely used for the design and optimization of CMOS technology, integrated circuits, and high-speed, low-power semiconductor devices.



This image presents the FETToy simulation of a Single-Gate MOSFET, showing the Transconductance Efficiency (gm/Id) versus Gate Voltage (V_g). The graph compares two simulation curves, illustrating that the gm/Id ratio decreases as the gate voltage increases, reflecting the variation in transistor efficiency across different operating regions. A higher gm/Id value at lower gate voltages indicates better transconductance efficiency, making the device suitable for low-power operation. This simulation is widely used to evaluate and optimize MOSFET performance for analog circuits, CMOS technology, integrated circuits, and energy-efficient nanoelectronic applications.

CONCLUSION: The MOSFET scaling characteristics were successfully analyzed using the nanoHUB FETToy simulation tool. Three important characteristics were studied: I_d – V_g , I_d – V_d , and gm/Id – V_g . The I_d – V_g graph shows that drain current increases with gate voltage, indicating effective channel formation. The I_d – V_d graph shows that drain current increases

with drain voltage and reaches a saturation region, while higher gate voltages produce higher drain current. The g_m/I_d - V_g graph shows higher transconductance efficiency at lower gate voltages, which is useful for low-power operation. Comparing all three graphs, they collectively explain the gate control, output characteristics, saturation behavior, and power efficiency of the MOSFET. Thus, the experiment provides a clear understanding of MOSFET operation and its importance in CMOS, low-power, and nanoscale electronic devices.